

25.09.2026 items review

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September 2026

1 Status of the 25.09 items

1. **Fairness. Done.** The checkpoint is built into the solver: before accepting a link, a matching check makes sure every UE can still be served, so every UE is served in every slot. Post processing became a validity check. Fairness across slots was added: proportional fair (PF) weights on good links only (SINR within 3 dB of the UE's best link), using a moving average of each UE's rate.
2. **Classical Greedy. Done.** Greedy colouring adapted from Appendix A (Algorithm 2, arXiv:2301.02637). Nodes are (UE, AP) links, colours are timeslots, exactly T colours are used and every colour covers every UE. All runs are valid ($K = 10 \times 5$ and $K = 15 \times 3$, seed 2). PF on good links keeps 98.5–99.7% of the efficiency and improves the worst served UE.
3. **QUBO/MIS formulation. Done.** Primary link MIS: same AP and same UE edges, a coverage bonus W and a penalty α . Verified without a solver: energies match the formula, and an exhaustive check on two clusters (32,768 and 4,096 states) gives the best covering assignment as the ground state.

2 Context

1. **Quantum is not yet useful.** At $N = 1$ with coverage, each slot is close to a bipartite matching, which classical methods solve exactly. The solver only has an advantage once pilot contamination or interference constraints are added. *Next: identify a realistic problem.*

2. **Interference as a cost.** Add a “cost if both” term for links that interfere when they are active in the same slot:

$$E(x) = - \sum_n w'_n x_n + \sum_{(n,m) \in H} \alpha_{nm} x_n x_m + \sum_{(n,m)} \beta_{nm} x_n x_m.$$

3. **Capacity beyond one.** The most realistic representation possible, means more antennas per AP ($N > 1$): an AP can then serve several UEs per

slot, which may also give a better slot allocation. The current link graph only expresses $N = 1$ with pairwise edges. With antenna copies the QUBO grows and large scenarios become costly. Supporting $N > 1$ properly means changing the graph construction again (point 7).

4. **Weight precision.** Weak links differ by about a millionth, while the coefficients sit around 3.5–5.9. Before compressing the weights, measure how precise we need to be, and how precise the quantum hardware can be:

- Add a random $\pm 5\%$ to each weight, solve the QUBO again and check whether the best solution changes.
- Local tuning: set different scenarios for the same QUBO and check whether Q captures those changes.

5. **Embedding.** To review with the team next week, including how to qualify an embedding.

6. **Post-processing.** A mathematical proof is needed that the classical fill step after the sampler does not affect solution quality.

7. **Alternative formulation for $N > 1$.** Nodes are UEs and colours are $(AP, \text{timeslot})$ pairs. The antenna count becomes a capacity limit per colour, and a cost for handovers avoids unnecessary AP changes between slots. *To be explored next week with Wesley and the Pasqal team + brain storming to find alternative representations.*

3 Items

1. Install the required software on the cluster and run the pipeline on the emulator (switching to cloud compute is a one line change). **Due:** week 41
2. Weight precision study ($\pm 5\%$ perturbations).
3. Add the interference cost term β_{nm} to the QUBO.
4. Review embedding options with the team and define how to qualify an embedding. **Due:** week 41
5. Proof that post processing does not affect solution quality.
6. Identify a realistic problem with the team (pilot contamination, interference, $N > 1$). **Due:** week 41
7. Explore the graph reformulation for $N > 1$ (nodes = UEs, colours = $(AP, \text{timeslot})$) with Wesley and the Pasqal team. **Due:** week 41

4 Benchmarking plan

Same scenario and random seed used on every solver; every schedule is checked with the same validity rules.

- Classical Greedy (Appendix A): the baseline.
- QUBO/MIS, solved classically (checks the QUBO formulation agrees with Greedy).
- Pulser based quantum sampling on the emulator/QPU

Metrics: total spectral efficiency, coverage rate (must be 100%), worst UE rate, good link coverage, handovers, runtime against graph size.